

ARTIFICIAL INTELLIGENCE

ORIGINAL ARTICLE

Development of a Novel Artificial Intelligence Clinical Decision Support System for Aesthetic Surgery: AURA

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Abstract

Background: Aesthetic surgery requires integration of specialized knowledge with clinical expertise, yet traditional literature search methods are time-intensive when addressing specific clinical questions. While artificial intelligence has transformed various aspects of healthcare delivery, no AI clinical decision support systems have been specifically developed to enhance evidence-based practice in aesthetic surgery.

Objectives: We aimed to develop and evaluate AURA (Aesthetic surgery Using Retrieval Augmentation), a novel AI-powered clinical decision support system designed specifically for evidence-based guidance in aesthetic surgery.

Methods: AURA integrates a comprehensive database of 6,546 full-text open-access aesthetic surgery publications (January 2001-September 2024) with a commercial large language model using retrieval-augmented generation technology. System performance was rigorously assessed across 14 complex clinical scenarios spanning facial rejuvenation, body contouring, breast procedures, and general aesthetic surgery considerations. Evaluation metrics included faithfulness to source materials (0-1), answer relevancy (0-1), G-Eval correctness (0-1), SEM semantic quality scores and SEM confidence ratings.

Results: AURA demonstrated exceptional performance with mean scores of 0.94 for faithfulness, 0.86 for answer relevancy, and 0.77 for factual correctness. Semantic evaluation revealed strong results with average scores of 0.73 (SEM Score) and 0.80 (SEM Max Similarity), predominantly with moderate confidence ratings. Performance was notably stronger for established techniques and safety considerations compared to emerging procedures requiring comparative analysis.

Conclusions: We present AURA, the first specialized AI clinical decision support system for aesthetic surgery. This novel system effectively delivers relevant, accurate information across diverse aesthetic surgery domains based on peer-reviewed literature. AURA offers aesthetic surgeons an efficient, evidence-based resource for clinical decision support. Future development should focus on expanding knowledge sources and prospective clinical validation, with implementation emphasizing transparent source presentation to complement surgical expertise.

Aesthetic surgery encompasses a diverse array of procedures requiring nuanced clinical decision-making based on both established evidence and emerging techniques¹. Aesthetic surgeons stay current with rapidly evolving literature while tailoring approaches to individual patient needs and

1 anatomical variations². The expanding volume of research, diversity of techniques, and variability
2 in outcome reporting create significant knowledge management challenges, at times requiring
3 expert consensus, in aesthetic practice³.

4 Recent advances in artificial intelligence (AI), particularly large language models (LLMs), have
5 demonstrated considerable potential for supporting clinical decision-making across various
6 medical specialties⁴⁻⁶. However, LLMs alone face significant limitations when applied to
7 specialized surgical fields^{7,8}. These include knowledge cutoff dates, potential for fabrication of
8 information ("hallucinations"), and lack of domain-specific expertise critical for aesthetic surgery
9 applications^{7,9}. Another recent innovation in computer sciences and AI, Retrieval-augmented
10 generation (RAG) system, offers to address these limitations by combining the reasoning
11 capabilities of LLMs with targeted retrieval from specialized knowledge bases^{10,11}.

12 This study presents the development and evaluation of a novel RAG system, we named AURA
13 (Aesthetic surgery Using Retrieval Augmentation), specifically designed for clinical decision
14 support in aesthetic surgery. By integrating a comprehensive corpus of aesthetic surgery literature
15 with a commercially available LLM, our system aims to provide accurate, relevant, and
16 contextually appropriate responses to complex clinical questions encountered in aesthetic surgical
17 practice. We hypothesized that this approach would yield high performance across multiple
18 computational evaluation metrics while providing meaningful clinical decision support.

19 20 **METHODS**

21 **Document Library Creation**

22 We created a specialized aesthetic surgery literature database by searching PubMed Central for
23 publications from January 1, 2001, to September 1, 2024. Search and data fetching were done with
24 a specialized coding script developed by the study authors. Our search included the following
25 comprehensive keywords; "cosmetic surgery", "aesthetic surgery", "facial rejuvenation",
26 "rhinoplasty", "breast augmentation", "liposuction", "body contouring", "abdominoplasty",
27 "blepharoplasty", "facelift", "botox", "fillers", "breast reduction", "gynecomastia surgery", "non-
28 surgical aesthetic procedures", "laser skin resurfacing", "hair loss treatment", and "cosmetic facial
29 surgery". This search yielded 6,546 full-text open-access publications, all of which were included
30 in our knowledge base.

The document processing involved three key steps: (1) extracting text from various document formats, (2) standardizing formatting and removing non-informative content, and (3) dividing documents into manageable chunks of 500 characters with 50-character overlap to preserve clinical context while enabling precise information retrieval. Figure 1 depicts an overview of the document library creation and AI model workflow.

System Architecture

We programmed AURA to combine a specialized aesthetic surgery knowledge base with advanced AI technology through a user-friendly web interface. The system works in three main stages:

1. Knowledge Base Creation: All 6,546 articles were converted into a searchable database where similar medical concepts are automatically grouped together using advanced text analysis.
2. Query Processing: When a surgeon asks a question, the system analyzes the query to understand what information is needed, searches the knowledge base for the most relevant content, and retrieves the top 3 most relevant passages from the literature.
3. Response Generation: An LLM, OpenAI's o1-mini (OpenAI, San Francisco, CA), receives the retrieved literature passages as context and creates a comprehensive answer based only on this evidence, providing citations to the original sources¹².

The system presents answers through a simple web interface that shows both the response and the source material used to generate it, allowing surgeons to verify information and explore the primary literature when needed.

AI Model Integration

We selected OpenAI's o1-mini language model based on its previously demonstrated balance between performance and computational efficiency with reasoning¹². The AI integration included specialized instructions for interpreting medical terminology accurately, maintaining proper citation of source documents, and acknowledging uncertainty in areas with limited evidence. The system was configured with response parameters that balance comprehensive answers for established techniques with appropriate uncertainty acknowledgment for emerging procedures. Figure 2 illustrates the complete question handling process with AI model integration.

Evaluation Methodology

System performance was evaluated using 14 clinically relevant questions representing diverse domains of aesthetic surgery practice. These questions were designed to assess real-world clinical

scenarios that aesthetic surgeons encounter, covering facial procedures (rhinoplasty, blepharoplasty, facelifting), body contouring (abdominoplasty, liposuction), breast surgery (augmentation, reconstruction), and general considerations in aesthetic practice.

We employed multiple evaluation metrics to assess different aspects of system performance listed below.

1. Faithfulness (0-1): How accurately responses reflected information from retrieved documents¹³
2. Answer Relevancy (0-1): How pertinent responses were to the questions asked¹⁴
3. G-Eval Correctness (0-1): Factual accuracy based on current standards of practice^{15,16}
4. Semantic Quality Scores: Overall semantic quality (SEM Score) and maximum similarity to reference materials (SEM Max Similarity)^{17,18}
5. SEM Confidence Ratings: Qualitative assessment (low, moderate, high) of the system's certainty¹⁷

Clinical Evaluation Questions

The 14 evaluation questions were distributed across four major aesthetic surgery domains: facial rejuvenation (7 questions covering rhinoplasty approaches, facial filler safety, tear trough management, combined rejuvenation approaches, fat grafting techniques, skin resurfacing comparisons, and SMAS preservation), body contouring (4 questions addressing abdominoplasty complications and buttock fat grafting safety), breast procedures (1 question on capsular contracture management), and general assessment (1 questions on body dysmorphic disorder screening and 1 question on minimum age recommendations for procedures). Each question was carefully crafted by the study authors to represent clinically relevant scenarios, ensuring diversity in complexity (e.g., technical approaches, safety, patient selection) and applicability to real-world practice while acknowledging that each be considered distinct sub-specialties, potentially warranting 10-20 questions per domain for exhaustive validation. Table 1 includes the list of clinical evaluation questions.

RESULTS

We have successfully developed the AI model. The evaluation of our AI system across 14 clinical questions demonstrated strong performance across multiple assessment dimensions. Overall, the system achieved an average faithfulness score of 0.94 (range: 0.67-1.00), indicating high

adherence to the retrieved context with minimal fabrication of information. Answer relevancy averaged 0.86 (range: 0.57-0.96), suggesting that responses successfully addressed the clinical questions posed. The G-Eval Correctness score averaged 0.77 (range: 0.60-1.00), reflecting good but not perfect factual accuracy. Semantic evaluation metrics showed average scores of 0.73 for SEM Score and 0.80 for SEM Max Similarity, with predominantly moderate confidence ratings. A complete breakdown of performance metrics by question is provided in Table 2. Figure 2 illustrates the complete question handling process. The full list of clinical questions and the system's responses are provided in the Appendix. Supplemental Figure 1 and Supplemental Figure 2 show the user interface of our system.

Performance varied across clinical domains as expected. Questions about established techniques with well-documented evidence bases received the highest scores. For example, questions related to post-abdominoplasty seroma formation (Q7, faithfulness: 0.97, relevancy: 1.00, correctness: 1.00) and SMAS preservation methods in facial procedures (Q11, faithfulness: 1.00, relevancy: 0.95, correctness: 0.80) achieved strong results. Facial rejuvenation queries showed consistent performance overall, likely reflecting the abundance of high-volume literature in this domain within the knowledge base.

In contrast, more modest performance was observed for questions concerning emerging techniques or those requiring comparative analysis. Questions regarding long-term outcomes of buttock fat grafting (Q8, faithfulness: 0.89, relevancy: 0.57, correctness: 0.60) and comparing microneedling versus laser resurfacing (Q9, faithfulness: 0.83, relevancy: 0.66, correctness: 0.60) received lower scores. This performance difference likely reflects greater variation in clinical approaches and evolving evidence in these areas with subjective and unvalidated outcome measurements utilized in many studies. The system appropriately acknowledged limitations in available evidence for these questions, with lower confidence ratings (both rated as "low") and more qualified conclusions.

Analysis of response content by clinical topic revealed that questions focusing on complication management and safety considerations received the most comprehensive and accurate answers. Questions on body dysmorphic disorder screening (Q6) and post-abdominoplasty seroma formation (Q7) both achieved perfect correctness scores of 1.00. Technical procedure descriptions also scored well, while patient selection criteria and outcome prediction questions showed more moderate performance. This pattern suggests the system's knowledge base and retrieval

mechanisms were particularly effective for safety-critical information and established technical approaches, areas where evidence-based guidance is most essential for clinical practice. The system demonstrated high faithfulness (>0.95) for 10 of the 14 questions, indicating strong adherence to the retrieved context in most clinical scenarios. Answer relevancy showed greater variation but remained above 0.75 for 11 of the 14 questions, suggesting effective retrieval and synthesis for most queries. Semantic confidence was predominantly rated as moderate (11 questions), with two questions rated low and one rated high, suggesting appropriate caution in the system's assertions for most clinical questions.

DISCUSSION

Our novel AURA system for aesthetic surgery demonstrated robust performance across multiple evaluation metrics, suggesting significant potential as a clinical decision support tool for practicing aesthetic surgeons. The high faithfulness scores (average 0.94) indicate that the system effectively leverages retrieved documents to generate responses with minimal fabrication of information. This characteristic is particularly crucial in aesthetic surgery and healthcare in general, where inaccurate information could directly impact surgical planning, patient outcomes and lead to quality and safety issues^{19,20}. Like previous examples in the literature, RAG models offer to provide a level of reliability that can exceed traditional LLMs operating without retrieval augmentation, addressing a critical need for evidence-based decision support in healthcare and aesthetic surgery by grounding responses in peer-reviewed literature^{21,22}.

The integration of a specialized knowledge base with advanced language modeling represents a significant innovation in clinical decision support for aesthetic surgery. Unlike general LLMs and AI systems, AURA's focused knowledge base of 6,546 papers ensures that responses are derived from domain-specific literature rather than general knowledge. This specialization allows for more precise and contextually appropriate answers to complex questions that arise in aesthetic surgical practice. The system's ability to maintain high answer relevancy (average 0.86) demonstrates its effectiveness in identifying and synthesizing pertinent information from this specialized knowledge base, potentially reducing the time surgeons spend searching through literature for specific clinical questions.

From a clinical application perspective, our AI model shows particular strength in addressing technical questions related to established surgical techniques. For example, questions regarding

1 post-abdominoplasty seroma formation, where methods to reduce seromas are well elucidated,
2 achieved near-perfect scores across all metrics (faithfulness: 0.97, relevancy: 1.00, correctness:
3 1.00), indicating reliable performance for procedural knowledge retrieval. Similarly, questions on
4 SMAS preservation methods in facelift surgery, despite a less definitive evidence base compared
5 to seroma management, scored highly (faithfulness: 1.00, relevancy: 0.95, correctness: 0.80),
6 demonstrating AURA's ability to provide accurate guidance even for techniques with ongoing
7 debate, excepting well-established aspects like hematoma reduction. This capability could be
8 immediately valuable for surgical planning and decision-making. By providing rapid access to
9 evidence-based technical approaches, the system could help standardize techniques and support
10 surgeons in implementing best practices, particularly those in training or early practice who may
11 benefit most from structured guidance on established and evolving procedures.

12 The system's strong performance on safety-related questions highlights its potential role in risk
13 management and complication prevention. Questions focusing on complication management and
14 patient safety considerations received the most comprehensive and accurate answers (average
15 correctness score of 0.83). This suggests that AURA could serve as an effective tool for pre-
16 operative planning and risk assessment, helping surgeons anticipate potential complications and
17 implement evidence-based preventive measures. In a specialty where patient satisfaction and
18 safety are paramount, having immediate access to current safety protocols and complication
19 management strategies could significantly enhance quality of care and reduce adverse outcomes.

20 For emerging techniques and those with evolving evidence bases, such as buttock fat grafting and
21 comparative facial rejuvenation approaches, the system's more modest performance reflects the
22 challenges inherent in areas with limited consensus. This limitation actually represents a strength
23 of the system, its ability to appropriately acknowledge uncertainty rather than providing false
24 confidence in areas where the evidence is still evolving. By maintaining moderate confidence
25 ratings for most questions and lower ratings for emerging techniques, AURA demonstrates
26 appropriate epistemic humility, a critical feature for clinical decision support tools that must
27 navigate areas of medical uncertainty without overstepping the bounds of available evidence²³.

28 The implementation of AURA in clinical practice could transform how aesthetic surgeons access
29 and utilize evidence during patient consultations, treatment planning, and surgical preparation.
30 With its transparent presentation of source materials alongside generated responses, the system
31 enables surgeons to verify information and explore primary literature when needed. This

transparency builds trust and supports appropriate use of the system as a complement to clinical judgment. The web-based interface allows for integration into existing clinical workflows, potentially at points of care where evidence-based decision support is most valuable, such as during preoperative planning sessions or patient consultations about procedure options and risks¹⁰. Future development of AURA could expand its capabilities through integration with emerging technologies and additional data sources. Incorporating structured data from electronic health records could enable more personalized recommendations based on patient-specific factors²⁴. Adding visual recognition capabilities for a multimodal model could allow the system to analyze pre-operative images and suggest appropriate surgical approaches based on anatomical variations²¹. Regular updates to the knowledge base would ensure the system remains current with evolving evidence and techniques. Additionally, implementing user feedback mechanisms would enable continuous refinement of the system based on real-world clinical utility and accuracy assessments from practicing surgeons.

Several important limitations warrant consideration. Our knowledge base, while comprehensive with 6,546 publications, encompasses only open-access literature, potentially missing recent publications, proprietary techniques, or high-impact studies behind paywalls that may contain superior evidence. The 24-year timespan, while providing sufficient volume for system development, includes potentially obsolete content that may influence responses inappropriately. The evaluation focused on 14 questions which, though methodologically justified, may not fully represent the breadth of clinical scenarios in aesthetic surgery practice, with uneven distribution across domains limiting generalizability of domain-specific conclusions. Each aesthetic surgery subspecialty warrants 10-20 questions for comprehensive validation, and our current question set serves as proof-of-concept requiring future expansion. The system provides information support but cannot replace clinical judgment, surgical expertise, or hands-on experience essential for aesthetic surgery practice. Evaluation metrics may not fully capture all aspects of clinical utility in real-world surgical contexts, and the study represents proof-of-concept work requiring prospective clinical validation, appropriate FDA regulatory pathways to assess impact on surgical planning and patient outcomes²⁵.

CONCLUSIONS

This study demonstrates successful development and validation of AURA, the first specialized AI clinical decision support system for aesthetic surgery. By combining comprehensive aesthetic surgery literature with advanced language modeling, AURA provides accurate, relevant clinical decision support across diverse practice domains. The system shows particular strength for established techniques and safety considerations while appropriately acknowledging limitations for emerging procedures. Implementation should emphasize transparent source presentation, critical evaluation of outputs, and integration as a complement to surgical expertise. Future development should focus on expanding knowledge sources beyond open-access limitations, enhancing retrieval for complex comparative questions, and conducting prospective clinical validation.

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FIGURE LEGENDS

Figure 1: AURA Workflow. Three-tier workflow diagram of the AURA (Aesthetic surgery Using Retrieval Augmentation) system showing: 1) Document Processing - utilizing PubMed API to search and filter 6,546 articles from 2001-2024 with aesthetic surgery keywords; 2) Knowledge Organization - creating hierarchical embeddings and maintaining a continuously updated vectorstore database with semantic indexing through LangChain and Chroma; and 3) Query Handling - processing user clinical questions through multiple LLMs to generate evidence-based clinical responses with citations.

Figure 2: Question Answering: Detailed View. Flowchart illustrating the detailed question processing pipeline of AURA, showing: input question processing by an initial LLM that generates sub-queries ($n=5$), semantic search retrieving relevant documents ($k=3$ per sub-query), context generation, and final response synthesis through a sequence of LLM operations. The output is presented through a user interface showing the answer, and retrieval context.

Table 1. Clinical questions to evaluate the developed AI model, AURA.

<i>Question #</i>	<i>Clinical Question</i>	<i>Domain</i>
<i>Q1</i>	What are the current best practices for managing capsular contracture following breast augmentation, and how do they differ between textured and smooth implants?	Breast Procedures
<i>Q2</i>	Compare the efficacy and safety profiles of different types of facial fillers (hyaluronic acid, calcium hydroxylapatite, poly-L-lactic acid) for midface volumization.	Facial Rejuvenation
<i>Q3</i>	What are the evidence-based guidelines for combining surgical and non-surgical facial rejuvenation procedures in a single session?	Facial Rejuvenation
<i>Q4</i>	Describe the anatomical considerations and technical approaches for addressing age-related changes in the periorbital region, specifically tear trough deformities.	Facial Rejuvenation
<i>Q5</i>	What are the latest advances in fat grafting techniques for facial rejuvenation, and how do processing methods impact graft retention?	Facial Rejuvenation
<i>Q6</i>	What preoperative assessment protocols best predict patients at risk for body dysmorphic disorder, and how should surgeons approach these cases?	General Assessment
<i>Q7</i>	What are the current evidence-based protocols for preventing and managing post-abdominoplasty seroma formation?	Body Contouring
<i>Q8</i>	What are the long-term outcomes of large-volume fat grafting to the buttocks, and what safety protocols have been developed to minimize fatal complications like fat embolism syndrome?	Body Contouring
<i>Q9</i>	Compare the efficacy of radiofrequency microneedling versus ablative laser resurfacing for facial skin rejuvenation in terms of downtime, complications, and long-term results.	Facial Rejuvenation
<i>Q10</i>	How does the use of progressive tension sutures in abdominoplasty affect seroma formation, pain management, and overall recovery time?	Body Contouring
<i>Q11</i>	What methods for preservation of the superficial musculoaponeurotic system (SMAS) have demonstrated superior outcomes in face and neck lifting procedures?	Facial Rejuvenation
<i>Q12</i>	What are the key anatomical landmarks for safe injection of fillers in the mid-face region?	Facial Rejuvenation
<i>Q13</i>	What is the recommended minimum age for cosmetic rhinoplasty in female versus male patients?	General Assessment
<i>Q14</i>	How does BMI affect patient selection criteria for abdominoplasty procedures?	Body Contouring

1 List of the 14 clinical questions used to evaluate AURA's performance across major aesthetic
2 surgery domains. Questions are categorized as follows: Facial Rejuvenation (7 questions: Q2, Q3,
3 Q4, Q5, Q9, Q11, Q12, Q13), Body Contouring (4 questions: Q7, Q8, Q10, Q14), Breast
4 Procedures (1 question: Q1), and General Assessment (2 questions: Q6, Q13).

5

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Table 2: Performance Metrics of AURA System Across 14 Clinical Questions

<i>Question</i>	<i>Faithfulness</i>	<i>Answer relevancy</i>	<i>G-Eval Correctness</i>	<i>SEM Score</i>	<i>SEM Max Similarity</i>	<i>SEM Confidence</i>
<i>Q1</i>	0.9000	0.9412	0.8000	0.8400	0.8700	moderate
<i>Q2</i>	1.0000	0.7497	0.8000	0.7700	0.8700	moderate
<i>Q3</i>	1.0000	0.8001	0.6000	0.6800	0.8200	moderate
<i>Q4</i>	1.0000	0.7979	0.6000	0.6600	0.7600	moderate
<i>Q5</i>	1.0000	0.8407	0.8000	0.6800	0.7800	moderate
<i>Q6</i>	0.9564	0.9346	1.0000	0.7600	0.8800	moderate
<i>Q7</i>	0.9668	0.9962	1.0000	0.7800	0.8600	moderate
<i>Q8</i>	0.8865	0.5672	0.6000	0.6400	0.7600	low
<i>Q9</i>	0.8328	0.6624	0.6000	0.6500	0.6700	low
<i>Q10</i>	1.0000	0.9283	0.8000	0.8600	0.9200	high
<i>Q11</i>	1.0000	0.9500	0.8000	0.8300	0.9000	moderate
<i>Q12</i>	1.0000	0.9340	1.0000	0.7000	0.7400	moderate
<i>Q13</i>	0.6666	0.9558	0.6000	0.7000	0.7200	moderate
<i>Q14</i>	1.0000	0.9320	0.8000	0.6600	0.6900	moderate
<i>Average Score</i>	0.9435	0.8564	0.7714	0.7293	0.8029	moderate

Summary of AURA's performance on each evaluation question across multiple assessment dimensions: Faithfulness to source documents (0-1), Answer Relevancy (0-1), G-Eval Correctness (0-1), Semantic Evaluation scores (SEM Score and SEM Max Similarity), and Confidence Ratings (High, Moderate, Low). The system achieved an average faithfulness score of 0.9435, answer relevancy of 0.8564, and correctness score of 0.7714 across all questions, with domain-specific performance variations.

AURA Workflow

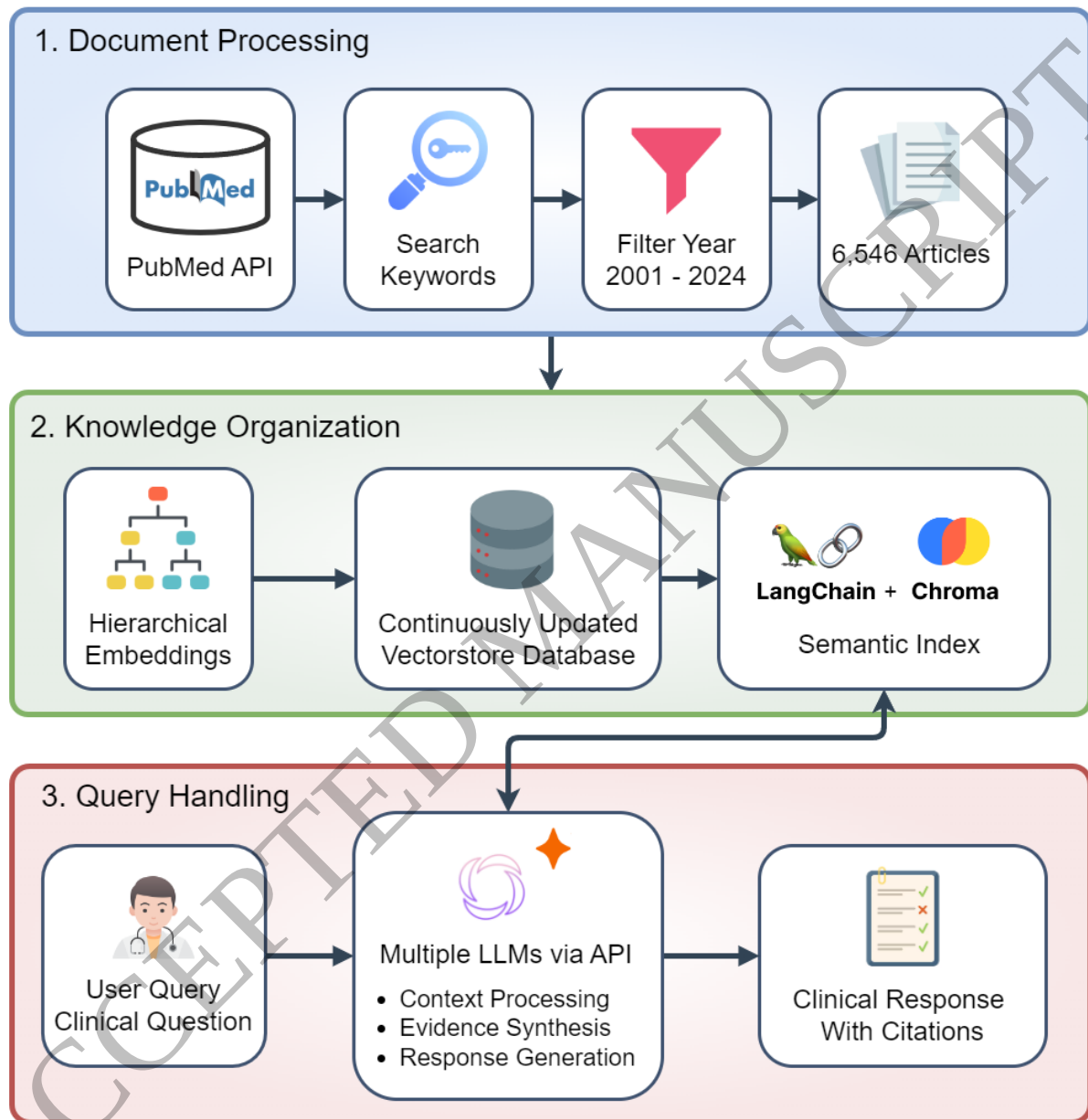


Figure 1
165x183 mm (x DPI)

Question Answering: Detailed View

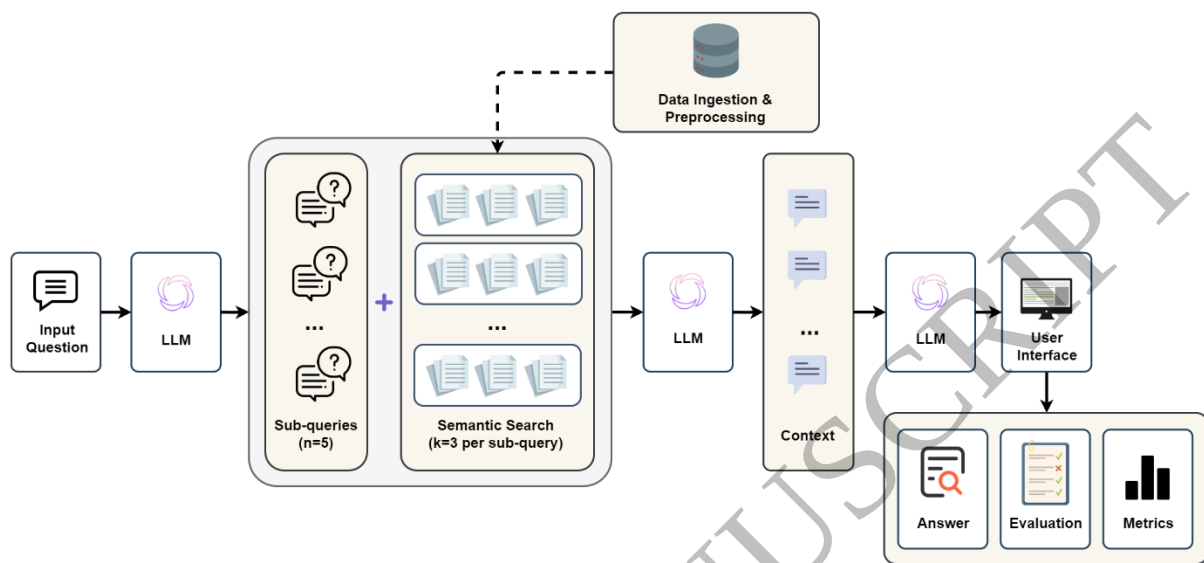


Figure 2
165x85 mm (x DPI)